⇒ Channelization Protocols :-

⇒ Channelization Protocol is a method used in computer n/ws to divide the communication channel into separate smaller channels. So that multiple devices can send data at the same time without interference. this helps in efficient use of bandwith and reduces data collisions

Multiplexing :- multiplexing in computer n/w means multiple signals are combined together thus travel at a time in a shared communication channel.

Types of Channelization Protocols (or) multiple Access Methods :-

① Frequency Division Mutiple Access (FDMA) :-

FDMA is a channel access methods used in various communication systems, allowing multiple users to share the same transmission medium by dividing the available bandwidth into different frequency bands. Each user is assigned a unique frequency bands. Each user is assigned a unique frequency band, so multiple devices can transmit data at a time without interference.

→ Key Characteristics of FDMA :-

1) Frequency Allocation :- The total available bandwidth is partitioned into multiple frequency channels. Each user is allocated a specific frequency band for communication. ensuring that transmissions do not cause collision.

2) Continuous Transmission :- Users can transmit data continuously within their assigned frequency band without waiting for time slot.

3) Guard Bands :- To prevent overlap and interference between adjacent frequency bands, small frequency gaps known as guard bands are implemented.

Applications of FDMA :-

1) Used in early mobile communication system, such as 1G mobile phones.

2) Used in FM radio stations.

Advantages of FDMA :-

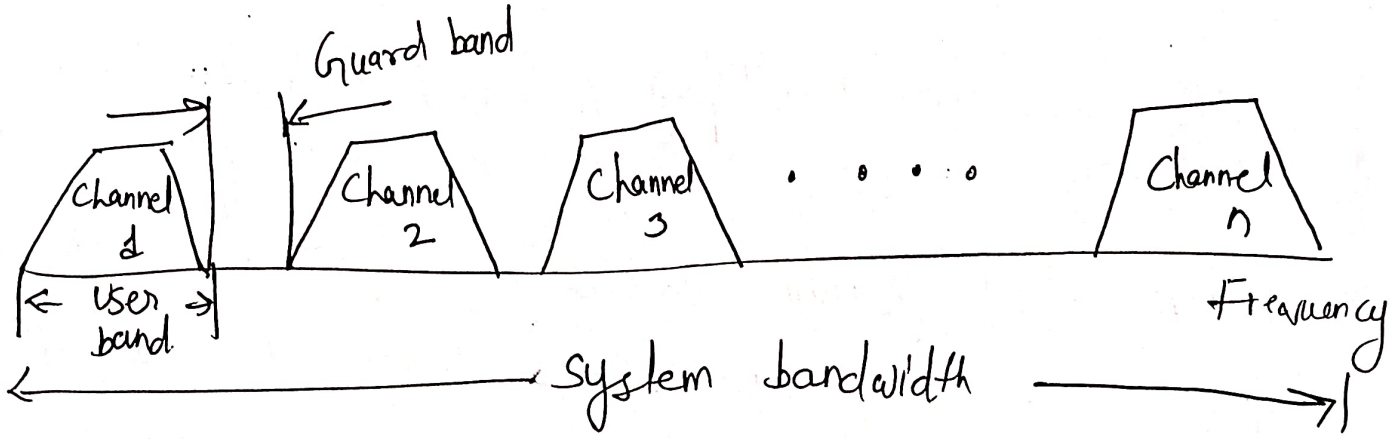
① Simplicity :- FDBM is slight forward to implement.

② Low Latency :- Since users have dedicated frequency bands, there is minimal delay in data transmission.

Disadvantages of FDMA :-

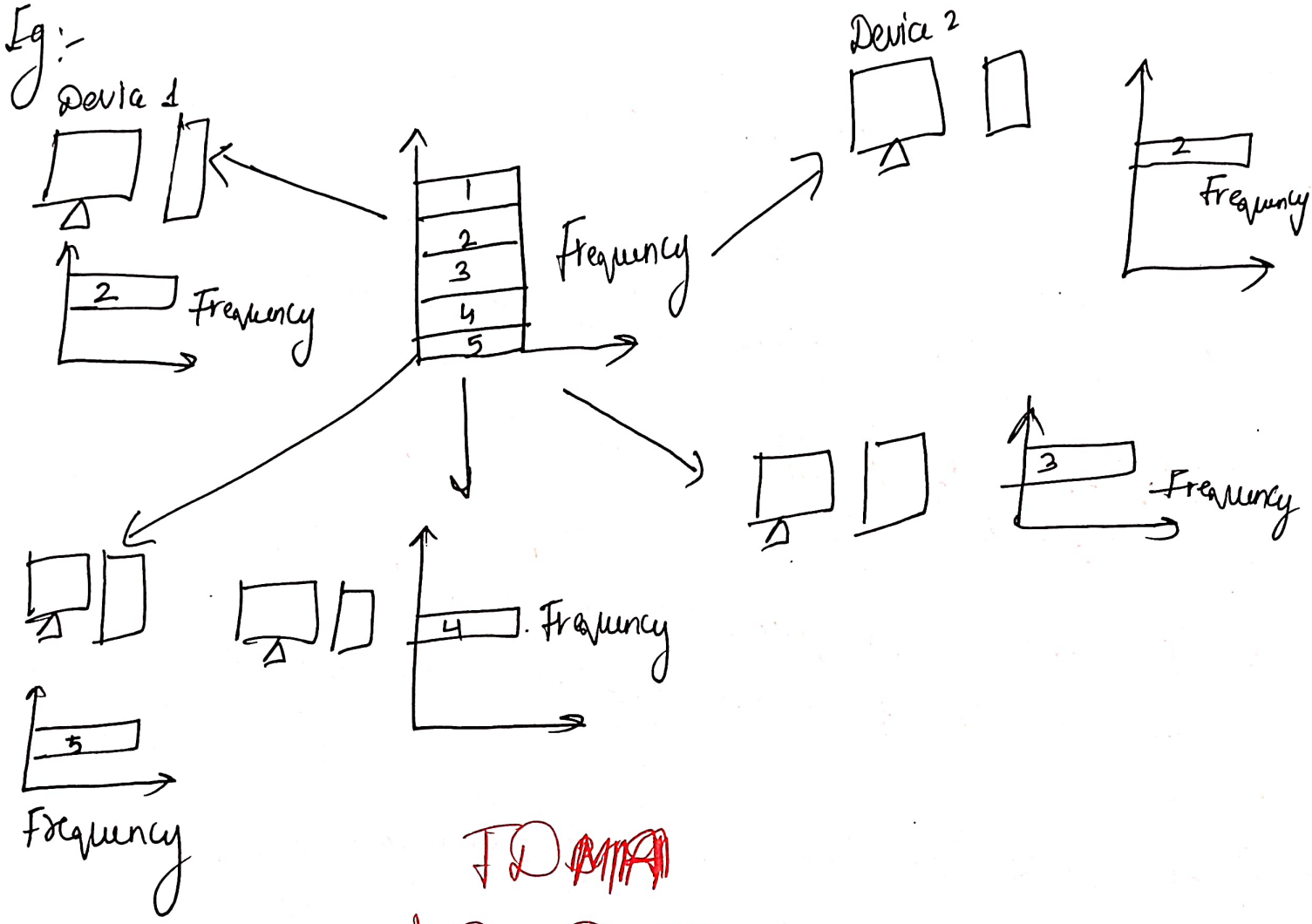
1) As fixed frequency is allocated to each user, if some users are inactive, resulting in under utilized frequency bands.

2) If users grow, hard to manage, frequency bands.



Communication Channel

Eg:-



TDMA

(2) Time Division multiple Access (TDMA):-

→ TDMA is a channel access method that enables multiple users to share the same frequency channel by dividing the signal into different time slots. Each user is assigned a specific time slot during which they can transmit or receive data, ensuring that multiple transmissions do not interfere with other.

→ Characteristics of TDMA :-

1) **Time slot allocation** :- The available bandwidth is segmented into time slots, with each user assigned a unique slot for communication.

2) **Synchronization** :- Each device must know exactly when its assigned time slot begins and ends, so that devices can not send data at the same time that could cause data collisions.

→ Advantages of TDMA :-

① **Efficient Bandwidth usage** :- By dividing the channel into time slots, multiple users can share the same frequency without interference.

② **Reduced Interference** :- Assigning unique time slots minimize the chance of overlapping signals, leading to clear communication.

③ Lower power consumption:- Devices transmit only during their allocated time slots, reduces power charges.

⇒ Disadvantages :-

1) users may experience delays waiting for their time slot, which can affect communication delay.

2) small gaps (guard times) b/w time slots are necessary to avoid interferences, slightly reducing data transmission capability.

3) following correct time is essential to prevent overlap b/w users.

⇒ 3) Code - Division Multiple Access (CDMA) :-

⇒ In CDMA multiple users share the same frequency band at a time by assigning each user a unique code. These distinct codes, enables the separation and identification of each users signal within the shared medium, allowing for efficient and secure communication

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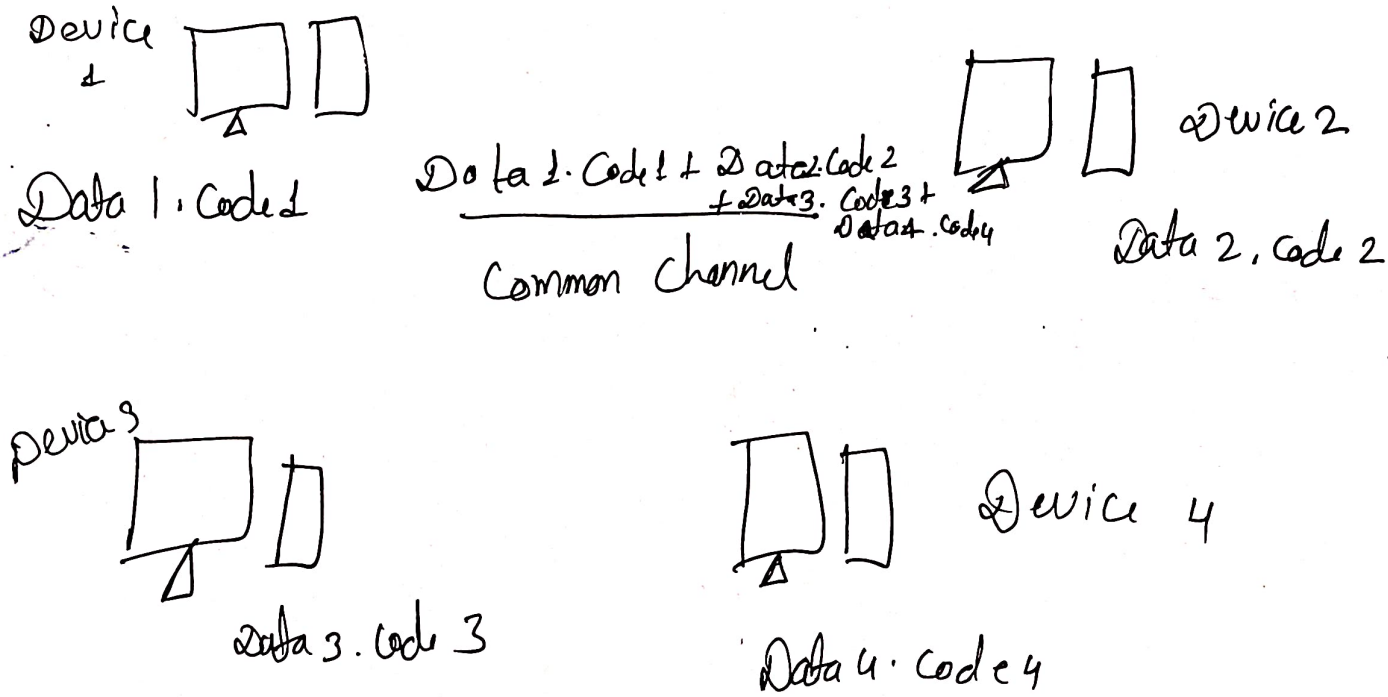
⇒ In CDMA, one channel carries all transmissions simultaneously Device 1.